

Lecture 2, Jan 23 - 2026.

Example: Find $(f^{-1})'(1)$ where $f(x) = 2x + \cos x$

Sln. Notice that

$$f'(x) = 2 - \sin x > 0$$

then, f is increasing and therefore f is one-to-one.

We know that

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} \quad \left\{ (f^{-1})'(1) = \frac{1}{f'(\underbrace{f^{-1}(1)})} \right.$$

We need $f^{-1}(1) = ?$

Notice $f^{-1}(y) = x \stackrel{\text{def}}{\Rightarrow} f(x) = y.$

Notice that $f(0) = 2(0) + \cos(0) = 1$ then $f^{-1}(1) = 0.$

then

$$(f^{-1})'(1) = \frac{1}{f'(f^{-1}(1))} = \frac{1}{f'(0)} = \frac{1}{2}.$$

6.2. Exponential functions and their Derivatives.

■ Exponential function.

In general, an exponential function is a function as.

$$f(x) = b^x$$

where b is a positive number.

If $x = n$, a positive integer, then

$$b^n = \underbrace{b \cdot b \cdot b \cdots b}_{n \text{ - times.}}$$

if $x=0$ then $b^0=1$ and $x=-n$ then

$$b^{-n} = \frac{1}{b^n}$$

if x is rational, $x = p/q$ where $p, q \in \mathbb{Z}$, $q > 0$ then

$$b^x = b^{p/q} = \sqrt[q]{b^p} = \left(\sqrt[q]{b}\right)^p$$

Thm. If $b > 0$ and $b \neq 1$ then $f(x) = b^x$ is a continuous with Domain $\mathbb{R}$ and range $(0, \infty)$. In particular, $b^x > 0$ for all $x \in \mathbb{R}$. If $0 < b < 1$, $f(x) = b^x$ is decreasing function; if $b > 1$, f is increasing function. If $a, b > 0$ and $x, y \in \mathbb{R}$ then:

i). $b^{x+y} = b^x b^y$

iii). $(b^x)^y = b^{xy}$

ii). $\frac{b^x}{b^y} = b^{x-y}$

iv). $(ab)^x = a^x b^x$

Prop. If $b > 1$ then $\lim_{x \rightarrow \infty} b^x = \infty$ and $\lim_{x \rightarrow -\infty} b^x = 0$

if $0 < b < 1$ then $\lim_{x \rightarrow \infty} b^x = 0$ and $\lim_{x \rightarrow -\infty} b^x = \infty$

Example. (a). Find $\lim_{x \rightarrow \infty} (2^{-x} - 1)$.

(b). sketch the graph of the function $y = 2^{-x} - 1$.

Sln. a).

$$\lim_{x \rightarrow \infty} (2^{-x} - 1) = \lim_{x \rightarrow \infty} 2^{-x} - \lim_{x \rightarrow \infty} 1$$

$$= \lim_{x \rightarrow \infty} 2^{-x} - 1$$

$$= \lim_{x \rightarrow \infty} \frac{1}{2^x} - 1$$

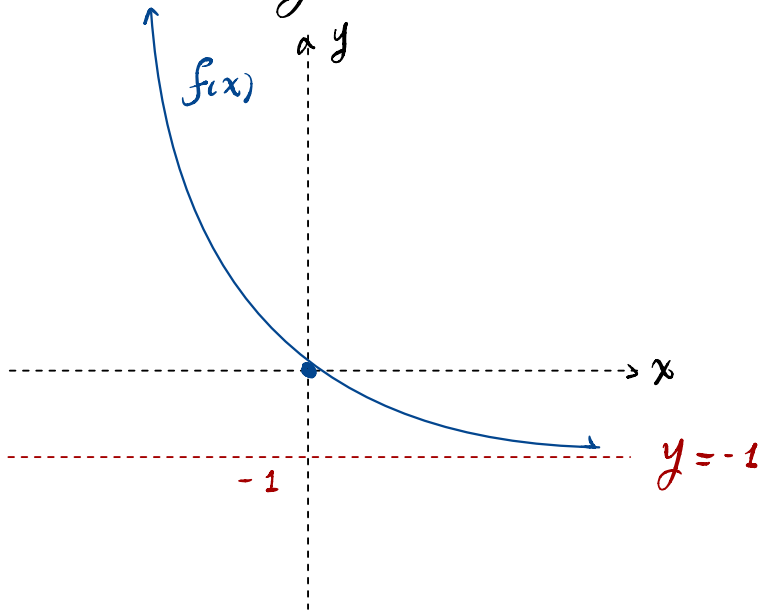
b/c $\lim_{x \rightarrow \infty} c = c$

$$\lim_{x \rightarrow \infty} (f(x) \pm g(x)) = \lim_{x \rightarrow \infty} f(x) \pm \lim_{x \rightarrow \infty} g(x)$$

$$= \lim_{x \rightarrow \infty} \underbrace{\left(\frac{1}{2}\right)^x}_0 - 1.$$

$$= 0 - 1 = -1.$$

(b). then $y = -1$ is H.A.



Derivatives and integrals of Exponential functions.

Let $f(x) = b^x$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{b^{x+h} - b^x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{b^x \cdot b^h - b^x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{b^x (b^h - 1)}{h}$$

$$= b^x \lim_{h \rightarrow 0} \frac{b^h - 1}{h} = b^x \cdot f'(0)$$

Thm. If $f(x) = b^x$ then

$$f'(x) = b^x \cdot f'(0).$$

Definition of the number e .

e is the number such that $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1.$ $\left\{ \right.$

Thm. $\frac{d}{dx}(e^x) = e^x$

Example. Differentiate the function $y = e^{\tan x}$.

Sln. $y' = e^{\tan x} \cdot \sec^2 x$.

prop:

$$\frac{d}{dx}(e^u) = e^u \cdot \frac{du}{dx}.$$

Example. Find y' if $y = e^{-4x} \sin 5x$.

Sln: Notice $y = f(x) \cdot g(x)$ where $f(x) = e^{-4x}$ $g(x) = \sin 5x$.

by the product rule

$$y' = f'(x)g(x) + f(x)g'(x).$$

$$\begin{cases} f'(x) = -4e^{-4x} \\ g'(x) = 5\cos 5x \end{cases}$$

$$\begin{aligned} y' &= -4e^{-4x} \sin 5x + 5e^{-4x} \cos 5x \\ &= e^{-4x} (5\cos 5x - 4\sin 5x). \end{aligned}$$

Example. (For you). Find the absolute maximum value of the function $f(x) = xe^{-x}$.

Thm: if $\frac{d}{dx}(e^x) = e^x$ then $\int e^x dx = e^x + C.$

Example. Evaluate $\int x^2 e^{x^3} dx.$

Sln. Let $f(x) = e^{x^3}$ let $u = x^3$ $du = 3x^2 dx.$
 $x^2 dx = \frac{1}{3} du$

by sub.

$$\int x^2 e^{x^3} dx = \int e^u \cdot \frac{du}{3} = \frac{1}{3} \int e^u du.$$

$$= \frac{1}{3} e^u + C.$$

$$= \frac{1}{3} e^{x^3} + C.$$

Example. Find the area under the curve $y = e^{-3x}$ from 0 to 1.

Sln. The area is

$$A = \int_0^1 e^{-3x} dx.$$

$$\int e^{-3x} dx = -\frac{1}{3} \int e^u du = -\frac{1}{3} e^{-3x} + C.$$

$u = -3x$
 $-\frac{1}{3} du = dx$

$$A = \int_0^1 e^{-3x} dx = -\frac{1}{3} e^{-3x} \Big|_0^1 = -\frac{1}{3} e^{-3} + \frac{1}{3} = \frac{1}{3} (1 - e^{-3}).$$